

AWS EBS (Elastic Block Store).

1. What is AWS EBS?

AWS EBS (Elastic Block Store) is a block-level storage service provided by Amazon Web Services that is used with Amazon EC2 instances.

Think of it like a virtual hard disk attached to a cloud server.

- It stores persistent data.
- Even if the EC2 instance stops or crashes, the data remains safe.
- Mainly used for databases, file systems, and applications that require high performance storage.

2. Key Characteristics

1. Block Storage

Data is stored in blocks (similar to SSD/HDD).

2. Persistent Storage

Data remains even after instance restart.

3. Attach/Detach Feature

EBS volumes can be attached or detached from EC2 instances.

4. Snapshot Support

Backup of EBS volumes can be stored in Amazon S3 as snapshots.

5. High Availability

Replicated automatically inside the same Availability Zone.

3. Types of EBS Volumes

Volume Type Description

Use Case

gp3 / gp2 General Purpose SSD Web servers, small databases io1 / io2

Provisioned IOPS SSD High-performance databases st1 Throughput

Optimized HDD Big data, analytics

Volume Type Description

Use Case

sc1

Cold HDD

Infrequent access

I have created an instance and then I have created an volume of 100 GB and attached to this instance.

The screenshot shows the AWS Management Console for EC2 Instances. The left sidebar contains navigation links for Instances, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays a table of instances. The 'linux-machine' instance (ID: i-0b7281389f6026b80) is selected and highlighted. Below the table, the details for this instance are shown, including its state (Running), public and private IP addresses, and DNS information.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
node exporter	i-068229086058279f4	Stopped	t3.micro	-	View alarms +	us-east-1a	-
loadbalancer	i-0024fc6fd21c29432	Stopped	t3.micro	-	View alarms +	us-east-1a	-
linux-machine	i-0b7281389f6026b80	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-100-54-
windows	i-0792d4709ebcdbc6	Stopped	t3.micro	-	View alarms +	us-east-1f	-

i-0b7281389f6026b80 (linux-machine)

Instance summary

Instance ID: i-0b7281389f6026b80

IPV6 address: -

Public IPv4 address: 100.54.39.236 [open address]

Private IPv4 addresses: 172.31.6.88

Instance state: Running

Public DNS: ec2-100-54-39-236.compute-1.amazonaws.com [open address]

Private IP DNS name (IPv4 only): -

You can check using the lsblk command.

```
[ec2-user@ip-172-31-6-88 ~]$ sudo su
[root@ip-172-31-6-88 ec2-user]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0   8G  0 disk /
|-nvme0n1p1 259:1    0   8G  0 part /
|-nvme0n1p127 259:2    0   1M  0 part /
|-nvme0n1p128 259:3    0   10M  0 part /boot/efi
nvme1n1     259:4    0  100G  0 disk
```

This is the created configuration storage of the new volume.

The screenshot shows the 'Create volume' page in the AWS Management Console. The 'Volume settings' section is expanded, showing the configuration for a new volume. The volume type is set to 'General Purpose SSD (gp3)'. The size is set to 100 GiB. The IOPS is set to 3000. The throughput is set to 125 MB/s. The availability zone is set to 'us-east-1a'. The snapshot ID is set to 'Don't create volume from a snapshot'. The encryption checkbox is unchecked.

Volume settings

Volume type: General Purpose SSD (gp3)

Size (GiB): 100

IOPS: 3000

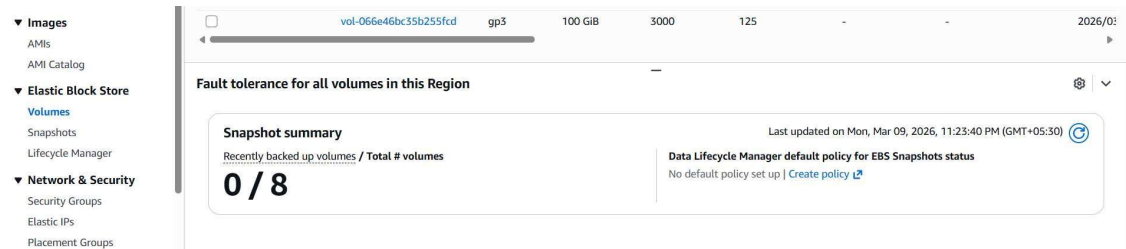
Throughput (MiB/s): 125

Availability Zone: use1-az1 (us-east-1a)

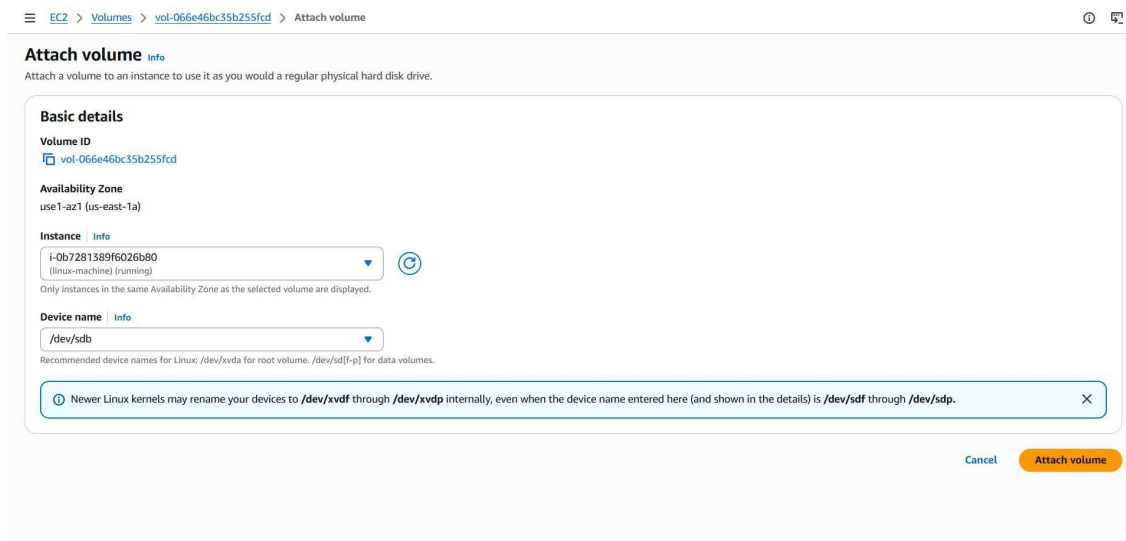
Snapshot ID - optional: Don't create volume from a snapshot

Encryption: ☐ Encrypt this volume

Created volume.



Then we have to attach to the instance but the EBS and EC2 have to be created in same region and same AZ's. Then we have to select the running ec2 machine.



When we use lsblk we can see the un mounted volume with free space.

```
[ec2-user@ip-172-31-6-88 ~]$ sudo su
[root@ip-172-31-6-88 ec2-user]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0   8G  0 disk 
├─nvme0n1p1 259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0  1M  0 part 
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1     259:4    0  100G  0 disk 
[root@ip-172-31-6-88 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           459M   0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1   8.0G  1.6G   6.4G  20% /
tmpfs           459M   0  459M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/1000
[root@ip-172-31-6-88 ec2-user]#
```

file -s /dev/nvme1n1 >>>> Checks if the disk already has a filesystem. mkfs -

t xfs /dev/nvme1n1 >>>> Creates a filesystem on the disk.

```
[root@ip-172-31-6-88 ec2-user]# mkdir /mnt/hari
[root@ip-172-31-6-88 ec2-user]# file -s /dev/nvme1n1
bash: file -s: command not found
[root@ip-172-31-6-88 ec2-user]# file -s /dev/nvme1n1
/dev/nvme1n1: data
[root@ip-172-31-6-88 ec2-user]# mkfs -t xfs /dev/nvme1n1
meta-data=/dev/nvme1n1      isize=512    agcount=16, agsize=1638400 blks
=                               sectsz=512    attr=2, projid32bit=1
=                               crc=1        finobt=1, sparse=1, rmapbt=0
=                               reflink=1    bigtime=1 inobtcount=1 nrext64=0
data      =                       bsize=4096   blocks=26214400, imaxpct=25
=                               sunit=1       swidth=1 blks
naming    -version 2          bsize=4096   ascii-ci=0, ftype=1, parent=0
log       -internal log      bsize=4096   blocks=16384, version=2
=                               sectsz=512    sunit=1 blks, lazy-count=1
realtime  -none              extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-6-88 ec2-user]#
```

i-0b7281389f6026b80 (linux-machine)
PublicIPs: 100.54.39.236 PrivateIPs: 172.31.6.88

mkdir /mnt/myvolume Creates a mount point directory where the volume will be attached.

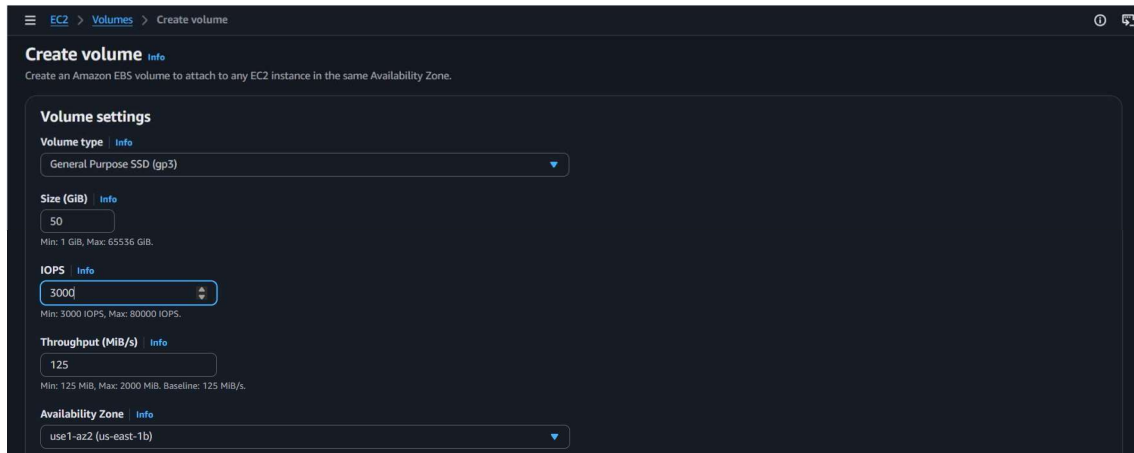
mount /dev/nvme1n1 /mnt/myvolume Mounts the EBS volume to the directory. df -h Shows disk usage and mounted filesystems.

```
[root@ip-172-31-6-88 ec2-user]# mount /dev/nvme1n1 /mnt/hari
[root@ip-172-31-6-88 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M  0  4.0M   0% /dev
tmpfs           459M  0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1  8.0G  1.6G  6.4G  20% /
tmpfs           459M  0  459M   0% /tmp
/dev/nvme0n1p2  10M  1.3M  8.7M  13% /boot/efi
tmpfs           92M  0  92M   0% /run/user/1000
/dev/nvme1n1    100G  747M  100G   1% /mnt/hari
[root@ip-172-31-6-88 ec2-user]#
```

i-0b7281389f6026b80 (linux-machine)
PublicIPs: 100.54.39.236 PrivateIPs: 172.31.6.88

EBS in Windows instance.

First we have to create an EBS in the same region and AZ.



Then we have to attach the volume to the created instance.

Successfully created volume vol-0ce8111eca07d641c.

Volumes (1/3) info

Last updated less than a minute ago

Recycle Bin Actions Create volume

Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Attach volume

Detach volume

Force detach volume

1

ID Created

vol-0ce8111eca07d641c

gp3

50 GiB

3000

125

vol-0120e9efa80c04d6d

gp3

30 GiB

3000

125

vol-0757ec798313501a9

gp3

30 GiB

3000

125

ID Created

2026/02

2026/02

2026/02

Basic details

Volume ID

vol-0ce8111eca07d641c

Availability Zone

use1-az2 (us-east-1b)

Instance info

i-0f8b91c2631ab263d (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name info

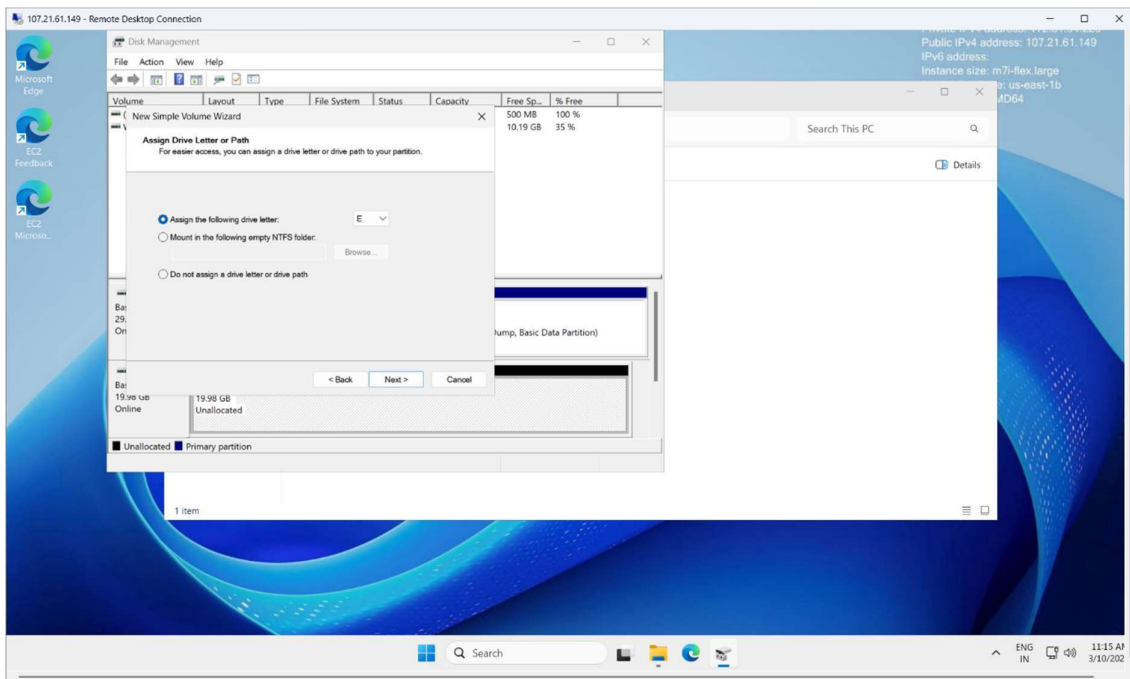
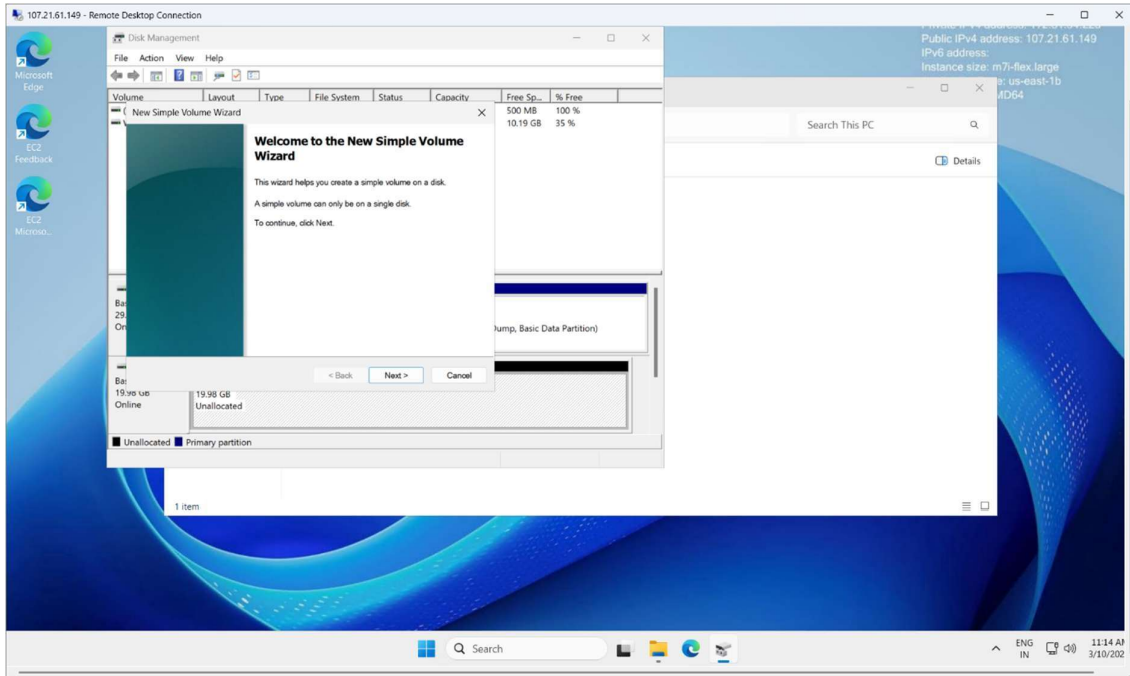
xvdb

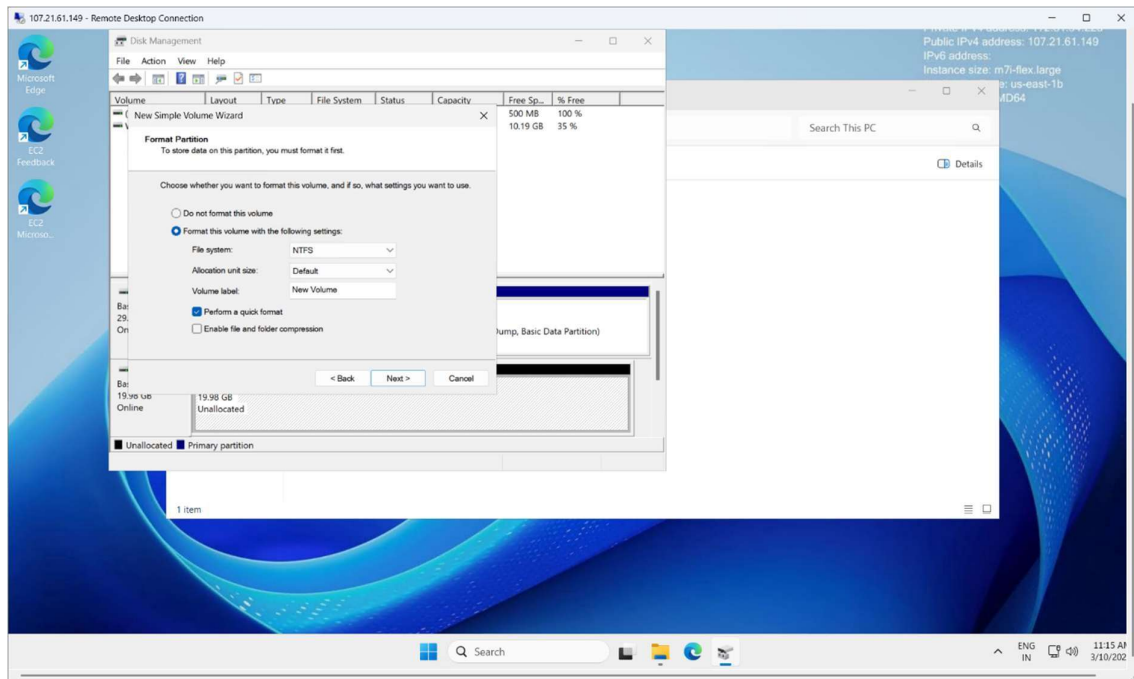
Recommended device names for Windows: /dev/sda1 for root volume, xvdf[-p] for data volumes.

Cancel

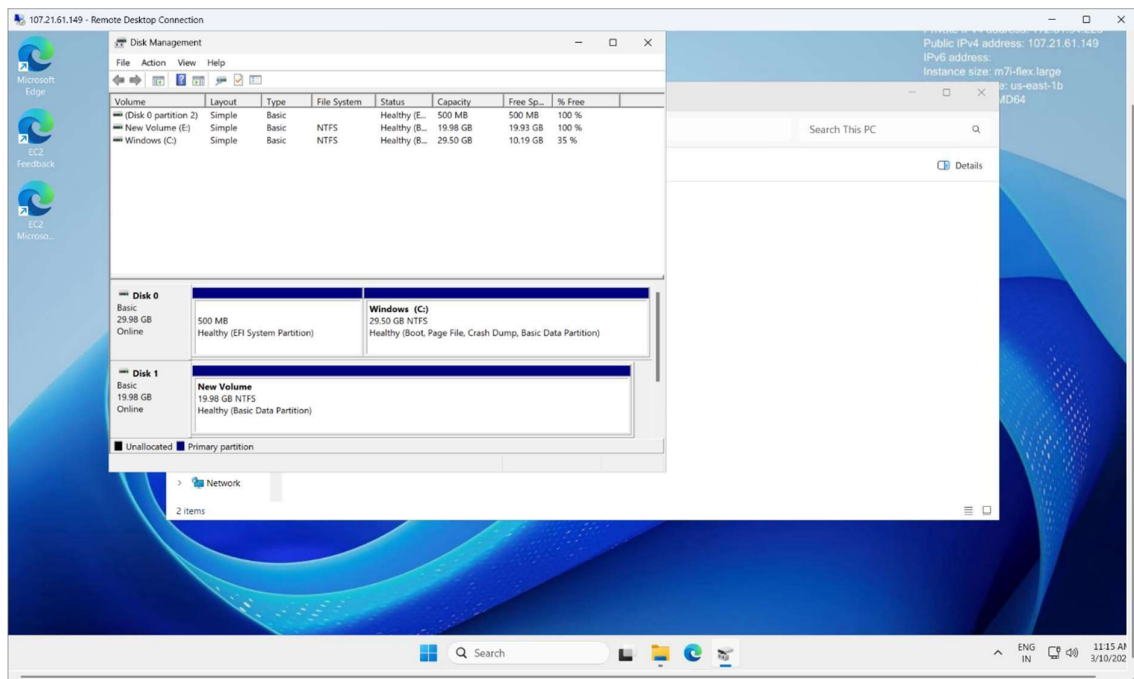
Attach volume

After searching for disk management you can see this app and you have to allocate the unallocated attached volume.

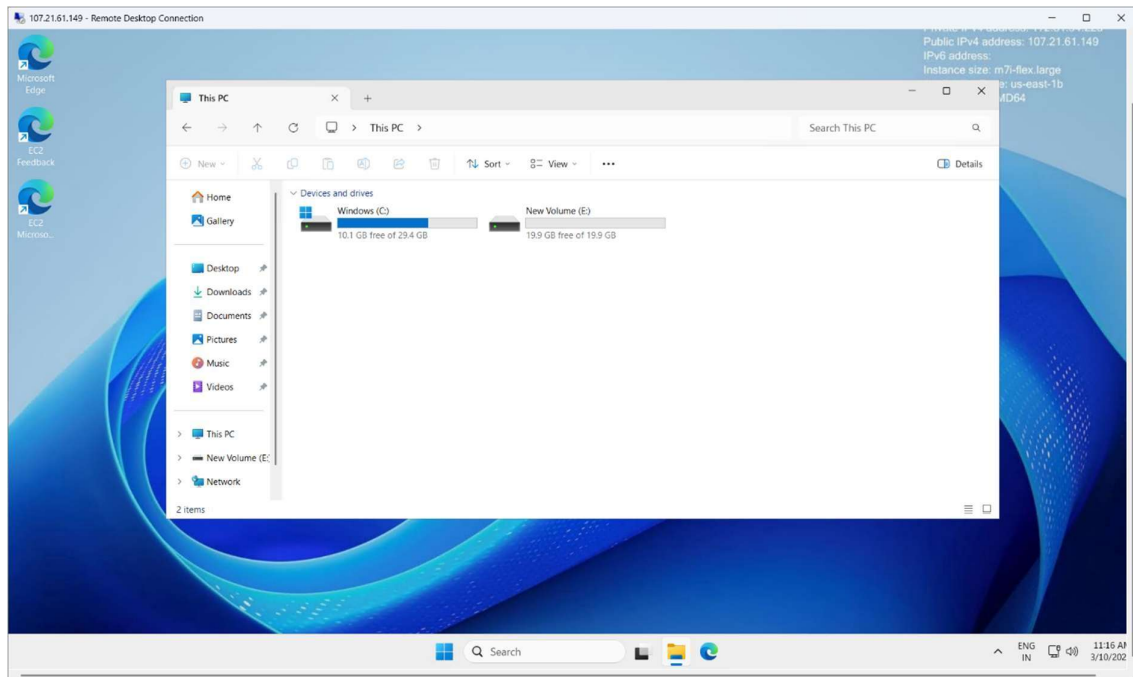




Then I have done all the procedure and the volume has been formatted and allocated.



We can see the allocated disk.



EBS Snapshot

An EBS Snapshot is a backup of an EBS volume stored in Amazon S3.

Key Points

- It is used to backup data of an EBS volume.
- Snapshots are incremental (only changes are saved after the first snapshot).
- Used for data recovery.
- You can create a new EBS volume from a snapshot.

Example Flow

1. EC2 instance has an EBS volume.
2. You create a snapshot of that volume.
3. Later you can restore it → create a new EBS volume → attach it to an instance.

I have selected the volume and creating one snapshot.

Volumes (1/2) Info

Name	Volume ID	Type	Size	IOPS	Throughput
☒	vol-08d319529cfb23313	gp3	20 GiB	3000	125
☐	vol-06ba02cd29b4f912b	gp3	30 GiB	3000	125

Volume ID: vol-08d319529cfb23313

Details | Status checks | Monitoring | Tags

Volume ID vol-08d319529cfb23313	Size 20 GiB	Type gp3	Status check Okay
AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more	Volume state In-use	IOPS 3000	Throughput 125
Fast snapshot restored No	Availability Zone use1-az2 (us-east-1b)	Created Tue Mar 10 2026 16:41:15 GMT+0530 (India Standard Time)	Multi-Attach enabled No
Attached resources i-09856ea384bfbdbbf (windows-machine)	Outposts ARN -	Managed false	Operator -

Create snapshot Info

Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.

Source volume

Volume ID: vol-08d319529cfb23313

Availability Zone: use1-az2 (us-east-1b)

Snapshot details

Description
Add a description for your snapshot.
mysnapshot
255 characters maximum.

Encryption
Not encrypted

Tags Info
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.

[Add new tag](#)
You can add up to 50 tags.

[Cancel](#) [Create snapshot](#)

Then I have the volume back (i.e) snapshot's

Snapshots (1) Info

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status
☐	snap-01a376de1d3d1920a	54 MiB	20 GiB	mysnapshot	Standard	Completed

AMI (Amazon Machine Image)

An AMI is a complete template used to launch an EC2 instance.

It contains:

- OS (Linux / Windows)
- Application software
- Configuration
- Snapshots of attached volumes

Key Points

- Used to launch new EC2 instances.
- Contains snapshots of EBS volumes.
- Acts like a golden template.

Example Flow

1. Configure an EC2 instance.
2. Install software (Apache, Docker, etc.).
3. Create an AMI from that instance.
4. Launch multiple instances using that AMI.

For creation of the image we can use the pre-installed webserver and any other services are build to AMI. By clicking the create image button for the selected instance

The screenshot displays the AWS Management Console interface for the EC2 service. The left-hand navigation pane shows the 'Instances' section selected. The main content area shows a list of instances with one instance, 'i-09856ea384bfbdbbf', highlighted. The instance is in a 'Running' state. The 'Actions' dropdown menu is open, showing options like 'Instance diagnostics', 'Instance settings', 'Networking', 'Security', 'Image and templates', 'Storage', and 'Monitor and troubleshoot'. Below the instance list, the details for the selected instance are shown, including the 'Instance summary' section with fields for Instance ID, Instance state, Public IPv4 address, Private IPv4 addresses, Instance state, Public DNS, and Private IP DNS name.

Name	Instance ID	Instance state	Instance type	Status check	Alarm state
windows-mac...	i-09856ea384bfbdbbf	Running	m7i-flex.large	3/3 checks passed	View alarm

i-09856ea384bfbdbbf (windows-machine)

Instance summary

Instance ID: i-09856ea384bfbdbbf

IPv6 address: -

Hostname type: -

Public IPv4 address: 107.21.61.149 | [open address](#)

Instance state: Running

Private IP DNS name (IPv4 only): -

Private IPv4 addresses: 172.31.94.223

Public DNS: ec2-107-21-61-149.compute-1.amazonaws.com | [open address](#)

Then we have to name the MYAMI.

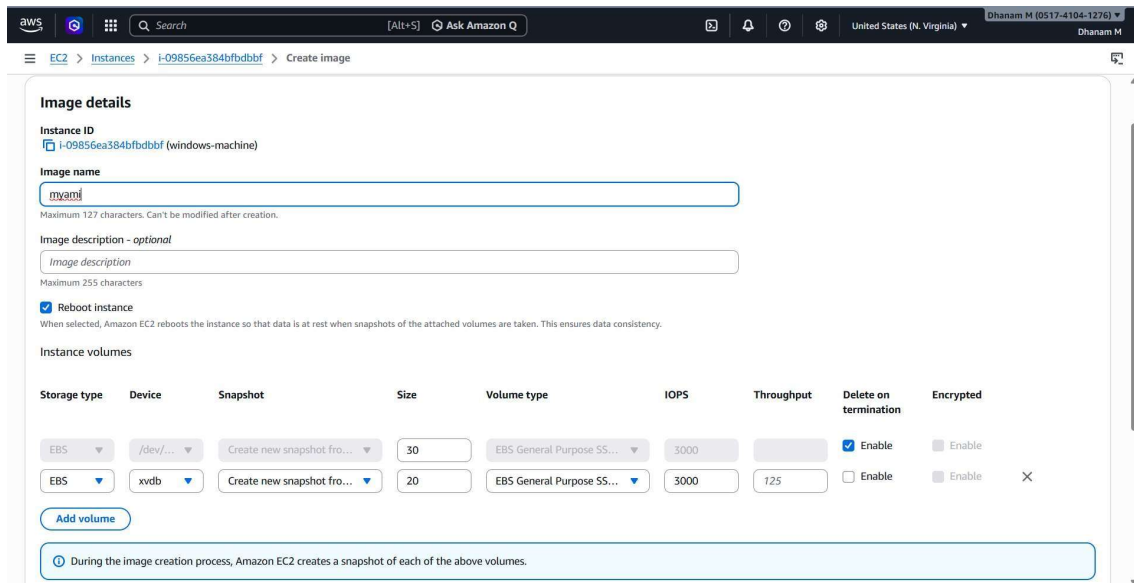


Image details

Instance ID
i-09856ea384bfdbdbf (windows-machine)

Image name

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Maximum 255 characters

☒ Reboot instance
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

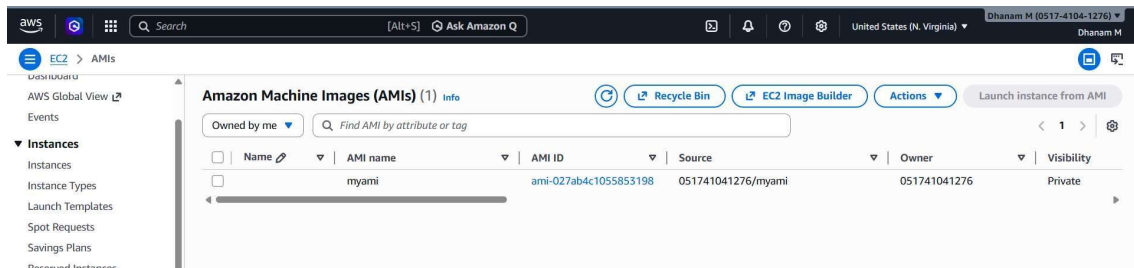
Instance volumes

Storage type	Device	Snapshot	Size	Volume type	IOPS	Throughput	Delete on termination	Encrypted
EBS	/dev/...	Create new snapshot from...	30	EBS General Purpose SS...	3000		☒ Enable	☐ Enable
EBS	xvdb	Create new snapshot from...	20	EBS General Purpose SS...	3000	125	☐ Enable	☐ Enable

[Add volume](#)

During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.

When the created MYAMI is available we can use it for launching an instance.

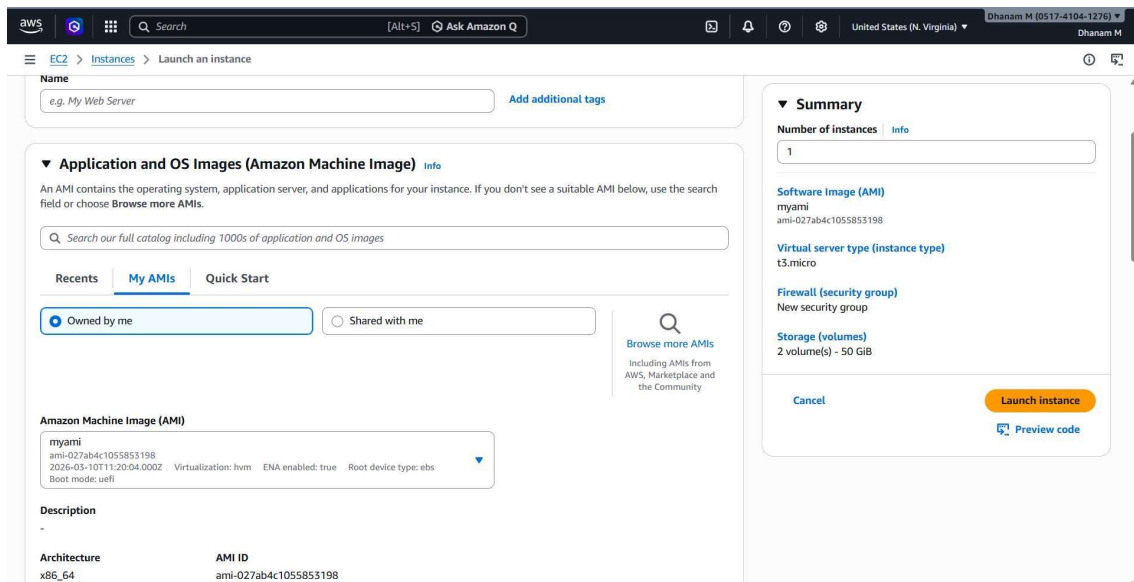


Amazon Machine Images (AMIs) (1)

Owned by me | Find AMI by attribute or tag

Name	AMI name	AMI ID	Source	Owner	Visibility
☐	myami	ami-027ab4c1055853198	051741041276/myami	051741041276	Private

The created MYAMI is in the created by me section.



Launch instance

Name

Application and OS Images (Amazon Machine Image)

Search our full catalog including 1000s of application and OS images

Owned by me | Shared with me

Amazon Machine Image (AMI)

myami
ami-027ab4c1055853198
2026-03-10T11:20:04.000Z
Virtualization: hvm
ENA enabled: true
Root device type: ebs
Boot mode: uefi

Description

Architecture	AMI ID
x86_64	ami-027ab4c1055853198

Summary

Number of instances: 1

Software image (AMI): myami (ami-027ab4c1055853198)

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 2 volume(s) - 50 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

VPC Peering.

1. What VPC Peering Actually Is

VPC Peering is a network connection between two Virtual Private Clouds (VPCs) that allows resources in both VPCs to communicate privately using private IP addresses.

No internet gateway.

No NAT traversal.

Traffic stays inside AWS network infrastructure.

In simple terms:

Two isolated networks become directly connected.

Example:

- VPC-A: 10.0.0.0/16
- VPC-B: 192.168.0.0/16

After peering, an EC2 instance in VPC-A can directly reach an EC2 in VPC-B.

2. Why VPC Peering Exists

AWS isolates VPC networks by default. Without peering, VPCs cannot communicate.

Peering is used when you want:

- Communication between multiple environments
- Communication between different AWS accounts
- Communication between different regions

Common cases:

- Dev VPC ↔ Production VPC
- Application VPC ↔ Database VPC

- Shared services VPC ↔ Multiple project VPCs

3. Types of VPC Peering

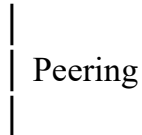
Intra-Region VPC Peering

Both VPCs are in the same AWS region.

Example

Mumbai region:

VPC-A (10.0.0.0/16)



VPC-B (172.16.0.0/16) Characteristics:

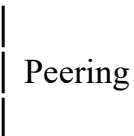
- Low latency
- No cross-region cost
- Simple routing

Inter-Region VPC Peering VPCs

are in different AWS regions.

Example:

VPC-A (Mumbai)



VPC-B (Singapore) AWS encrypts the traffic automatically.

Used for:

- Multi-region architectures
- Disaster recovery setups

I have created one VPC in the N.virginia region.

The screenshot shows the 'Create VPC' page in the AWS Management Console for the 'United States (N. Virginia)' region. The page is titled 'VPC settings' and includes the following configuration options:

- Resources to create:** 'VPC only' is selected.
- Name tag - optional:** 'myvpc' is entered.
- IPv4 CIDR block:** 'IPv4 CIDR manual input' is selected, and '10.0.0.0/16' is entered in the text field.
- IPv6 CIDR block:** 'No IPv6 CIDR block' is selected.
- Tenancy:** 'Default' is selected.
- VPC encryption control (\$):** 'None' is selected.

The other in the Mumbai region.

The screenshot shows the 'Create VPC' page in the AWS Management Console for the 'Asia Pacific (Mumbai)' region. The page is titled 'Create VPC' and includes the following configuration options:

- VPC settings:** 'VPC only' is selected.
- Name tag - optional:** 'myvpc1' is entered.
- IPv4 CIDR block:** 'IPv4 CIDR manual input' is selected, and '10.0.0.0/16' is entered in the text field.
- IPv6 CIDR block:** 'No IPv6 CIDR block' is selected.
- Tenancy:** 'Default' is selected.
- VPC encryption control (\$):** 'None' is selected.

And I have configured all the recommended settings for the VPC to Activate.

(i.e) internet gateway , security groups , subnets.

The creation of public instance in the N.virginia region.

The screenshot shows the 'Launch an instance' wizard in the AWS Management Console for the 'United States (N. Virginia)' region. The 'Key pair name' is set to 'hari'. The 'Network settings' section is expanded, showing the 'VPC' as 'vpc-084f2a0a522890979 (myvpc)' and the 'Subnet' as 'subnet-0f7fdb28678267d71' (public). The 'Auto-assign public IP' is set to 'Enable'. The 'Firewall (security groups)' section shows 'publicsg sg-023841209409f2738' selected. The 'Summary' section on the right shows 'Number of instances' as 1, 'Software image (AMI)' as 'Amazon Linux 2023 AMI', 'Virtual server type (instance type)' as 't3.micro', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. The 'Launch instance' button is visible.

The creation of public instance in the Mumbai region.

The screenshot shows the 'Launch an instance' wizard in the AWS Management Console for the 'Mumbai' region. The 'Network settings' section is expanded, showing the 'VPC' as 'vpc-084f2a0a522890979 (myvpc)' and the 'Subnet' as 'subnet-0f7fdb28678267d71' (public). The 'Auto-assign public IP' is set to 'Enable'. The 'Firewall (security groups)' section shows 'publicsg sg-023841209409f2738' selected. The 'Summary' section on the right shows 'Number of instances' as 1, 'Software image (AMI)' as 'Amazon Linux 2023 AMI', 'Virtual server type (instance type)' as 't3.micro', and 'Storage (volumes)' as '1 volume(s) - 8 GiB'. The 'Launch instance' button is visible.

Before peering we have to give one more route values to all the VPC's.

1st VPC.

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
11.0.0.0/16	Peering Connection	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

2nd VPC.

Edit routes

Destination	Target	Status	Propagated	Route Origin
11.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
10.0.0.0/16	Peering Connection	-	No	CreateRoute

Buttons: Add route, Cancel, Preview, Save changes

Then this is the Peering of the two VPC's From Different Regions.

This is the requesting form from the N.virginia Region.

Create peering connection

Select a local VPC to peer with

VPC ID (Requester): vpc-084f2a0a522890979 (myvpc)

VPC CIDRs for vpc-084f2a0a522890979 (myvpc)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-

Select another VPC to peer with

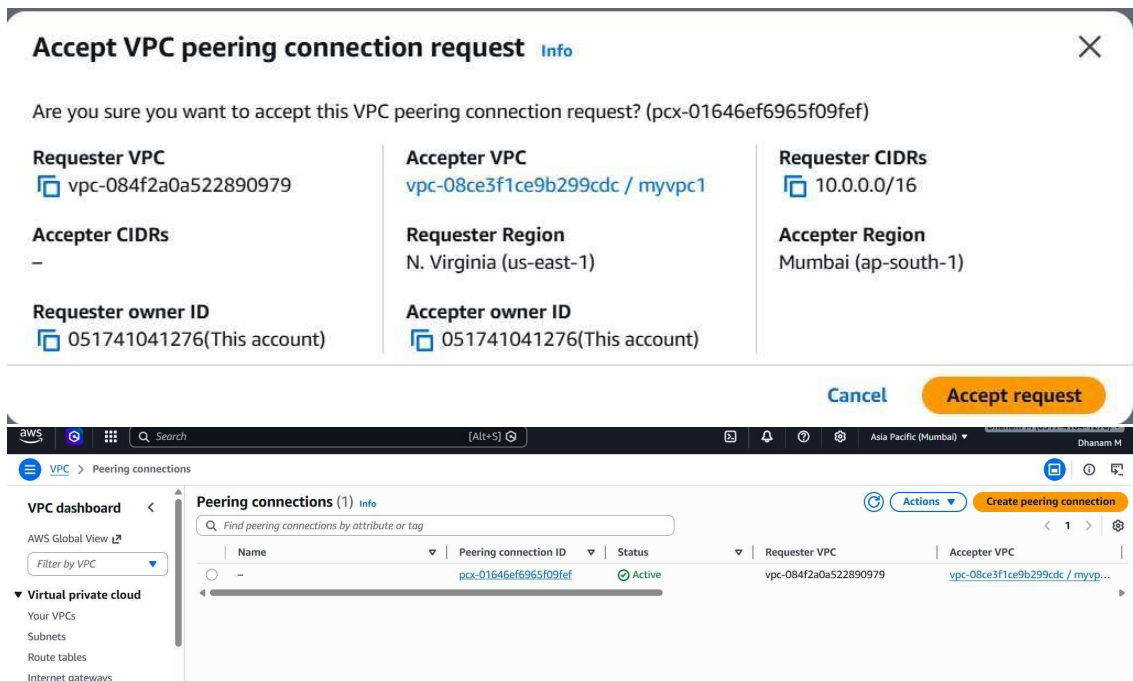
Account: ☒ My account ☐ Another account

Region: ☐ This Region (us-east-1) ☒ Another Region

Asia Pacific (Mumbai) (ap-south-1)

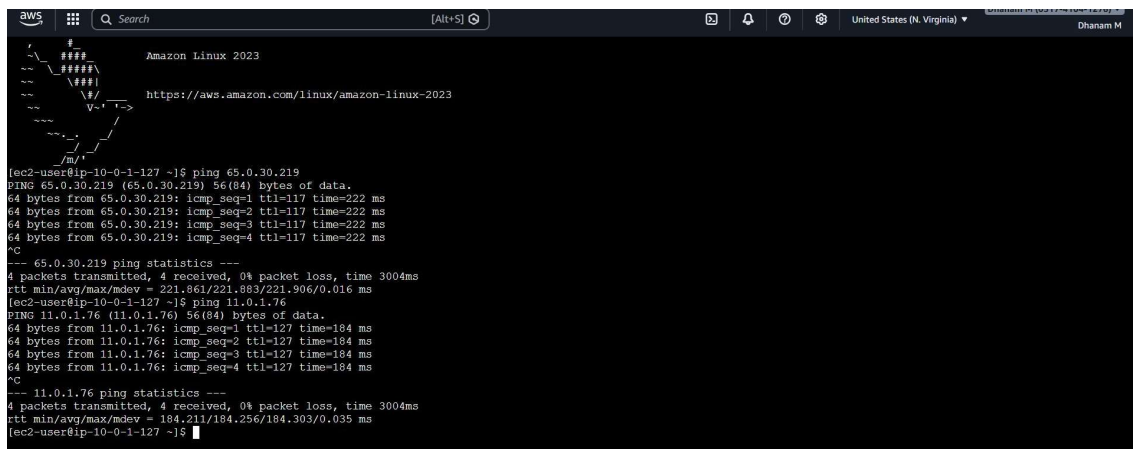
VPC ID (Acceptor): vpc-08ce3f1ce9b299cdc

This is the Acceptor form in the Mumbai Region.



We are using ping command to check whether the internet is connected with the two VPC.

1st Instance.



2nd Instance.

```
aws [Search] [Alt+S] Asia Pacific (Mumbai) Dhanam M

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-11-0-1-76 ~]$ ping 13.223.100.210
PING 13.223.100.210 (13.223.100.210) 56(84) bytes of data.
64 bytes from 13.223.100.210: icmp_seq=1 ttl=108 time=222 ms
64 bytes from 13.223.100.210: icmp_seq=2 ttl=108 time=222 ms
64 bytes from 13.223.100.210: icmp_seq=3 ttl=108 time=222 ms
64 bytes from 13.223.100.210: icmp_seq=4 ttl=108 time=222 ms
^C
--- 13.223.100.210 ping statistics ---
5 packets transmitted, 4 received, 20% packet loss, time 4005ms
rtt min/avg/max/mdev = 221.879/221.905/221.919/0.015 ms
[ec2-user@ip-11-0-1-76 ~]$ ping 10.0.1.127
PING 10.0.1.127 (10.0.1.127) 56(84) bytes of data.
64 bytes from 10.0.1.127: icmp_seq=1 ttl=127 time=184 ms
64 bytes from 10.0.1.127: icmp_seq=2 ttl=127 time=184 ms
64 bytes from 10.0.1.127: icmp_seq=3 ttl=127 time=184 ms
64 bytes from 10.0.1.127: icmp_seq=4 ttl=127 time=184 ms
^C
--- 10.0.1.127 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 184.231/184.259/184.317/0.034 ms
[ec2-user@ip-11-0-1-76 ~]$
```